

MDF-MLLM: Deep Fusion Through Cross-Modal Feature Alignment for Fundoscopy Classification

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Abstract—This study aimed to enhance disease classification accuracy from retinal fundus images by integrating fine-grained image features and global textual context using a novel multimodal deep learning architecture. Existing multimodal large language models (MLLMs) often struggle to capture low-level spatial details critical for diagnosing retinal diseases such as glaucoma, diabetic retinopathy, and retinitis pigmentosa. Our Multi-Depth Fusion (MDF-MLLM) model integrates skip features from four U-Net encoder layers into cross-attention blocks within a LLaMA 3.2 11B MLLM. Vision features are patch-wise projected and fused using scaled cross-attention and FiLM-based U-Net modulation. On a dual-type classification task, MDF-MLLM achieved an accuracy of 94%, while the baseline MLLM achieved 60% accuracy on the same task. When expanding the dual-type classification to all individual disease classes, the baseline MLLM achieved 26% overall accuracy whereas MDF-MLLM achieved 57%. Ablation studies confirmed that the multi-depth fusion approach contributed to substantial gains in spatial reasoning and classification, particularly for inherited diseases with rich clinical text. MDF-MLLM presents a generalizable, interpretable, and modular framework for fundus image classification, outperforming traditional MLLM baselines through multi-scale feature fusion. The proposed architecture holds promise for real-world deployment in clinical decision support systems.

Index Terms—Cross-modal fusion, Multimodal large language model, U-Net, Retinal fundus photography, Vision transformer.

I. INTRODUCTION

Ophthalmologists rely heavily on medical imaging to help discern key features within the human eye. Fundus photography is especially important, as it gives a detailed and colored view of the back of the eye, highlighting the structural components of the retina and its blood vessel pattern. The shape and structure of these components provide ophthalmologists and retina specialists with the necessary information for detecting abnormalities associated with eye diseases. The vascular structure of the retina is of particular importance, with retinal blood vessel shape, size, and damage all serving as important indicators of vision health, as well as overall cardiovascular health [1], [2]. Therefore, the development of diagnostic and data analysis tools that can provide a multi-depth understanding of local and global features to efficiently detect diagnostically relevant retinal structures would be very useful for ophthalmologists and retina specialists. The current work presents an architecture that enables the acquisition of multimodal (image and text)

biomedical knowledge and provides helpful insights for medical practitioners within clinical settings.

Recent advances in artificial intelligence (AI), machine learning (ML), multimedia, and large language model (LLM) research open new engineering possibilities for designing systems that can combine visual and textual data in a unified framework [3]–[7]. In particular, multimodal LLMs (MLLMs) offer a pathway toward more context-aware diagnostic support tools by enabling information fusion across data types. However, challenges remain in adapting general-purpose models to specialized clinical domains. This study presents an alternative solution by introducing MDF-MLLM (Multi-Depth Fusion of MLLM), a cross-modal integration architecture that combines fine-grained image features with high-level contextual embeddings to improve the classification capability of an open-source MLLM (LLaMA 3.2 11B Vision-Instruct) with appropriate reasoning capacity to assist ophthalmologists in medical imaging diagnostics. The MDF-MLLM architecture incorporates fusion methods designed to optimize the MLLM model for contextually aware image classification, which was validated on several open-source retinal fundus datasets (FIVES [8], HRF [9], and the Disease Atlas from StoneRounds.org [10]). The main contributions of the research presented here can be summarized as follows:

1. Demonstrated the capability of a multimodal LLM to integrate and optimize contextual visual information from individual U-Net model layers through enhanced fusion techniques that incorporate semantic and spatial alignment.
2. Assessed the effectiveness of facilitating transfer learning on model performance through supplementary fine-tuning on open-source and customized retinal fundus image datasets.
3. Validated the multimodal LLM’s ability to harness the appropriate contextual information and effectively enhance predictive capacity for disease category classification from retinal fundus images and patient clinical text information while presenting the response predictions in a clinically relevant fashion for effective reasoning and interpretability.

The remainder of this paper provides an overview of related work within the fields of ophthalmology and AI in Section II. Section III outlines the base model architecture and provides a theoretical framework for the proposed feature fusion methods that focuses on the enhancement of the multimodal

LLM functionality. Section IV describes the dataset structure, provides a description of the experimental setup with evaluation metrics for overall performance assessment, and highlights the key findings of the study. The concluding remarks in Section V discuss the unique contributions of this research and offer additional insights into future research opportunities.

II. RELATED WORK

In medicine and, more specifically, ophthalmology, AI/ML has progressed along three largely independent tracks: (1) pixel-level segmentation, which extracts the anatomic priors clinicians need; (2) large language models (LLMs) that parse text notes or generate narrative reports; and (3) vision-language fusion frameworks that promise end-to-end clinical reasoning but are still in their infancy for retinal care.

DeepVessel, learned hierarchical representations that delineated vessel boundaries and already exceeded handcrafted filters on benchmark datasets [11]. U-Net utilizes CNNs to extract low-level features from an image via a contracting encoder to capture context, followed by a symmetric expanding decoder path to enable precise localization, and introduced encoder-decoder skip connections that became a standard for medical image segmentation [12]. Many variants of this system have been developed and stem from variations in the base CNN. Deformable-U-Net (DU-Net), enriched U-Net with deformable convolutions to adaptively adjust for the vessels' shapes and sizes [13]. Dense Residual Vessel U-Net (DR-VNet) utilizes residual dense net modules along with squeeze-and-excitation blocks for improving the task with small-diameter blood vessels [14]. These systems were later expanded to incorporate attention mechanisms, as with Spatial Attention U-Net (SA-U-Net) [15] and Multi-Scale Multi-Attention Network (MSMA-Net) [16].

As LLMs gain traction in clinical medicine, their application in ophthalmology is emerging as a key area of interest, prompting early evaluations of their utility and risk [17]. A recent review on the integration of LLMs in ophthalmology identified more than 100 relevant studies, highlighting promising applications in triage, clinical documentation, and patient education [18]. For successful implementation, practical and ethical challenges must be addressed before LLMs can be reliably implemented in real-world clinical settings [19].

General-purpose vision-language pre-training offers mechanisms to close the previously mentioned gap surrounding the limitations of LLMs in highly specialized domains. BLIP-2 (Bootstrapping Language-Image Pre-training) bridges frozen image encoders and LLMs with a lightweight Querying Transformer [20]. Medical-specific adaptations of MLLM architectures have also emerged, such as Cross-Modal Augmented Transformer (CAT) which injects attention blocks between retinal encoders and language decoders to generate coherent diagnostic reports [21].

Despite these advances, several critical limitations remain. Most frameworks (e.g., BLIP-2) rely heavily on pretrained, general-purpose encoders, which often fail to capture fine-grained spatial or structural features such as vessel morphology

or micro-lesions that are crucial in ophthalmic diagnostics. Recent surveys have reported that current MLLMs often struggle to ground detailed visual attributes in the absence of explicit textual annotations, leading to poor sensitivity to subtle but clinically important findings [22].

Therefore, there is a clear gap in developing architectures that can simultaneously preserve low-level semantic precision from medical images and leverage the high-level contextual reasoning power of MLLMs in a unified model architecture. Our proposed MDF-MLLM addresses this by integrating U-Net-based multi-scale features with LLaMA's vision-language embeddings through multi-depth cross-attention fusion.

III. METHODS

The proposed MDF-MLLM architecture is shown in Fig. 1 and displays the base U-Net model interconnected with the LLaMA 3.2 Vision multimodal large language model (MLLM) through customized U-Net fusion and Cross-Attention fusion blocks. The U-Net encoder architecture incorporates ResNet50 (Residual Network) [23] modules for the first four convolutional layers and the bridge connection, with the initial layers each outputting a skip connection. Leveraging the pretrained CNN layers into the encoder architecture through transfer learning is intended to enhance the model's generalization capabilities, speed training, and reduce overfitting. These skip connections are intercepted and passed through the proposed modules, where they are fused with key layers along the MLLM pipeline.

The MLLM architecture is composed of a vision encoder and a language decoder. The language decoder contains 40 sequential layers that consist of 32 self-attention blocks and 8 cross-attention blocks. Given that U-Net skip connections represent a vision component of the overall fusion architecture, cross-modal fusion was implemented at the cross-attention layers (3, 18, 28, and 38) for each of the four U-Net encoder outputs. These specific layers were chosen to allow an even distribution of U-Net information to be broadcast throughout the language pipeline and to better match the level of feature depth in the latent space.

To evaluate the effectiveness of MDF-MLLM, input images were subjected to several preprocessing steps (detailed in Section III-A) for standardization and then trained across several configurations seeking to optimize diagnosis accuracy. Model training was conducted on a series of publicly available retinal fundus images (FIVES, HRF, and StoneRounds), and a portion of the image dataset also included anonymized patient information utilized by clinicians for health assessments. Together, these features were coordinated to optimize the MLLM's ability to effectively categorize patient health.

A. Image Preprocessing and Prompt Initialization

Retinal fundus images were presented to the model predominantly in native form with the intention of preserving the base features that will be made available to the MLLM. The only preliminary modifications involved resizing to $3 \times 512 \times 512$ ($C \times H \times W$) and normalization, which involved element-wise division by 255. Certain images also contained watermark

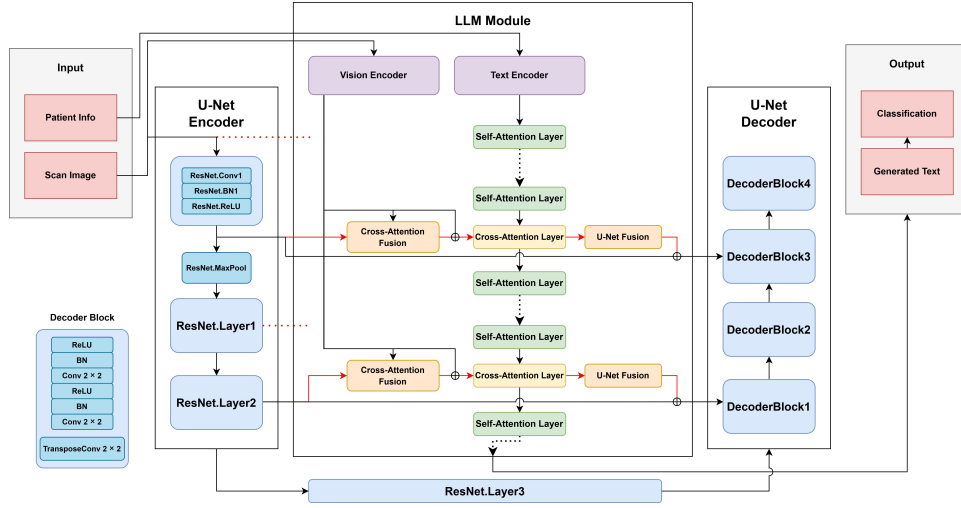


Fig. 1: Overall architecture of MDF-MLLM, showing U-Net model with ResNet50 encoder backbone integrated with LLaMA 3.2 11B Vision model cross-attention layers via customized fusion modules. Black arrows are existing connections of the U-Net and LLM model, while red arrows symbolize the connections made for new fusion components. Dotted lines symbolize replicated fusion block pipelines processed through additional MLLM text encoder layers, and have been omitted for clarity.

identifiers from the original publishers. Augmentation steps were applied to remove these logos, and included watermark region isolation, conversion of the selected region to grayscale, threshold application to create a binary mask, expanding the mask with a 3×3 dilation kernel, mask removal from the original image region, and finally filling in the subtracted pixel regions using a Fast Marching Method (FMM) inpainting algorithm which propagates information from the surrounding pixels that remain.

The MLLM architecture was fine-tuned to recognize features associated with varying disease types and to assess whether a specific fundus image belonged to a given disease class. To ensure consistent results, an instruction prompt was customized to guide the MLLM to act as an expert clinician and assess the health of a patient’s eye based on the provided fundus image and additional information, if available. Fig. 2 (top section) displays the generalized prompt used in instructing the model. The middle section of Fig. 2 displays examples of relevant patient-specific text and fundus images, with text being attached to the end of the generalized prompt, and the bottom section displays their respective ground truth labels. It should be noted that the images utilized from the public segmentation datasets (FIVES and HRF) did not contain additional patient information, other than disease classification. Therefore, these images were simply prompted with the phrase “This is a fundus photography image,” and the reference text only listed the specific disease type.

B. Cross-Attention Fusion

U-Net skip features were fused to the MLLM vision encoder outputs before integrating into the model’s cross-attention layers. For fusion implementation, we define k skip feature maps as $S^{(k)} \in \mathbb{R}^{C_k \times H_k \times W_k}$ that have C_k channels (starting with 3 channels at $k = 1$ for RGB images and transforming to

64, 256, 512 for $k = 2, 3, 4$, respectively) and spatial resolution $H_k \times W_k$ (512×512 for $k = 1$ and halving in each dimension for each increase in k). The feature map is divided into $p \times p$ non-overlapping patches (16×16 for the present study), then flattened into vectors as shown in Eqn. 1, with $N_k = \frac{H_k W_k}{p^2}$ representing the number of patches.

$$P^{(k)} = \text{reshape}(\text{unfold}_{p,p}(S^{(k)})) \in \mathbb{R}^{N_k \times C_k p^2} \quad (1)$$

Each patch vector is projected into the LLM embedding dimension d , which is 4096 on the LLaMA 3.2 11B Vision model, using a level-specific linear projection as shown in Eqn. 2, with trainable weights ($W_{pat}^{(k)}$) and bias ($b_{pat}^{(k)}$).

$$E^{(k)} = P^{(k)} W_{pat}^{(k)} + b_{pat}^{(k)} \in \mathbb{R}^{N_k \times d} \quad (2)$$

To demonstrate cross-attention fusion, let $V_L \in \mathbb{R}^{T \times d}$ be the MLLM embedded vision features, where T is the sequence length (representing the number of patches on the MLLM vision encoder), and $d = 4096$ is the hidden dimension of the MLLM embedded space. The MLLM vision features are fused with the U-Net encoder features using attention mechanisms by setting the query (Q), key (K), and value (V) parameters as shown in Eqn. 3, with $|\cdot|_2$ representing the Euclidean norm and W and b representing trainable parameters.

$$Q^{(k)} = \frac{V_L}{|V_L|_2} W_Q^{(k)} + b_Q^{(k)}, \quad K^{(k)} = \frac{E^{(k)}}{|E^{(k)}|_2} W_K^{(k)} + b_K^{(k)}, \\ V^{(k)} = E^{(k)} W_V^{(k)} + b_V^{(k)} \quad (3)$$

Finally, these projections were passed through a dot product attention layer followed by a normalization step, as shown in Fig. 3, to better ensure compatibility with the MLLM pipeline. The outputs are then scaled by α , which is a constant controlling the strength of the injected fusion signal, and then clamped

"You are a medical diagnostic assistant specializing in Ophthalmology. Analyze the provided retinal image and assess the patient's eye health. If any signs of diseases or abnormalities are present, provide a diagnosis or possible conditions. Be clear, concise, and use clinical terminology where appropriate."		GENERALIZED PROMPT
 This is a fundus photography image for the left eye of a 7-year-old female with a visual acuity of 20/20. This 7-year-old girl first noticed poor vision in dim light when she was three years old. She has no difficulties with photophobia or color vision.	 This is a fundus photography image.	PATIENT-SPECIFIC PROMPT
The diagnosis disease is AR Retinitis Pigmentosa, which is associated with the PDE6B gene, Met1Thr ATG>ACG - Leu489Pro CTG>CCG alleles, and AR inheritance mode.	The diagnosis disease is Diabetic Retinopathy	REFERENCE TEXT

Fig. 2: MLLM prompt layout with generalized prompt used on all data inputs (top), retinal fundus image with patient-specific information (center row), and reference text (bottom row) for model output comparison. The left column represents a patient with an inherited disease type, while the right column represents a patient with an acquired disease type.

to $(-5, 5)$ range before being passed through a trainable linear layer and a final layer norm projection.

Following feature fusion, the revised embeddings are summed to the original vision encoder parameters through a residual connection and are passed as inputs to the MLLM cross-attention layer for integration with the language embeddings. Fig. 3 provides a visual overview of the cross-attention fusion architecture.

C. U-Net Fusion

Within the base U-Net encoder architecture, each layer includes a series of CNN layers followed by Batch Normalization and a ReLU nonlinear activation. This CNN block is followed by a down-sampling layer that increases the number of feature channels and halves the number of spatial features. These blocks are repeated three additional times, effectively reducing a $512 \times 512 \times 3$ input image to $32 \times 32 \times 1024$, and a copy of the output from each ReLU operation is saved as a skip connection. These connections would normally be concatenated to a corresponding up-convolution block within the decoder architecture of a traditional U-Net structure; however, the proposed fusion method seeks to incorporate additional vision and language feature information from the MLLM cross-attention layers before passing on to the decoder.

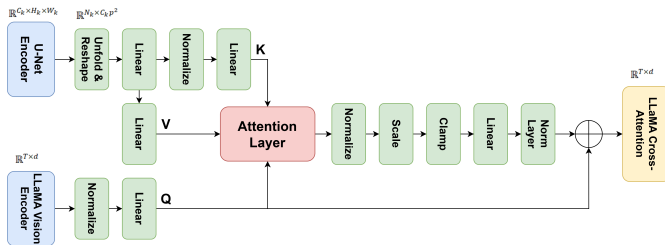


Fig. 3: Cross-Attention Fusion block for merging U-Net skip connections to LLaMA Vision embeddings.

It is important to ensure that the higher-level context injected into the skip connection features is spatially well-aligned and matches channel depth. To account for this, the U-Net fusion module incorporates a feature-wise linear modulation (FiLM) technique [24] in which MLLM cross-attention outputs (H) are normalized to unit length and mean-pooled across the sequence (axis T), as shown in Eqn. 4. $H \in \mathbb{R}^{T \times d}$ denotes the output of a cross-attention layer, where T is the sequence length and $d = 4096$ is the MLLM hidden dimension.

$$\bar{h} = \frac{1}{T} \sum_{t=1}^T \frac{H_{t,:}}{\|H_{t,:}\|_2} \in \mathbb{R}^d \quad (4)$$

Assuming $S^{(k)} \in \mathbb{R}^{C_k \times H_k \times W_k}$ is the feature map output of the image from the k level U-Net encoder, a channel-wise modulation vector is created for each skip map k , and an affine transformation is produced by:

$$\hat{S}_{c,y,x}^{(k)} = \gamma_c^{(k)} \cdot S_{c,y,x}^{(k)} + \beta_c^{(k)} \quad (5)$$

where $S_{c,y,x}^{(k)}$ denotes the activation in spatial location (y, x) in channel c , and $\gamma_c^{(k)}, \beta_c^{(k)} \in \mathbb{R}^{C_k}$ represent the linear projection slope and bias parameters (Eqn. 6), respectively. $\hat{S}^{(k)}$ will then be used as input to the decoder at level k .

$$\gamma^{(k)} = \bar{h} W_\gamma^{(k)} + b_\gamma^{(k)}, \quad \beta^{(k)} = \bar{h} W_\beta^{(k)} + b_\beta^{(k)} \quad (6)$$

IV. RESULTS AND DISCUSSION

A. Data Preparation and Evaluation

1) *Datasets*: Three primary data sources were used for this study. Two sources include the publicly available FIVES and HRF image datasets. FIVES (Fundus Image Vessel Segmentation) consists of 800 high-resolution fundus photographs with segmentation masks and is classified into labeled groups consisting of healthy patients and others diagnosed with either type 1 or 2 diabetes mellitus, age-related macular degeneration (ARMD), or glaucoma. The HRF (High Resolution Fundus) dataset contains 15 images of healthy, diabetic retinopathy (DR), and glaucomatous eye scans, and annotated masks.

The final dataset was extracted from the StoneRounds.org [25] and NEI Data Commons [26] repositories. These include 460 images of the most prevalent diseases, including AR Retinitis Pigmentosa (AR RP), AD Retinitis Pigmentosa (AD RP), XL Retinitis Pigmentosa (XL RP), Best Disease, and AR Stargardt Disease. Many StoneRound patient images also included clinical information that describes symptoms, family history, diagnosis reasoning, etc., which was utilized for fine-tuning the MLLM. FIVES and HRF datasets did not include text data. To complete the dataset, the vascular network within the selected StoneRounds images were professionally annotated for training with the U-Net architecture. Since the first two datasets lacked clinical text associated with the patient images, this research chose to exclude images associated with healthy patients since it represented only a single class and text is considered to contain key attributes to aid in model outputs and classification. Therefore, all results focused on distinguishing between diseased images grouped into either

acquired or inherited disease types with the intent to expand the classification categories in future works. The 1,090 image-text pairs were divided into sets of 70%, 10%, and 20% for training, validation, and testing, respectively.

2) *Evaluation Metrics*: To evaluate the performance of MDF-MLLM, classification accuracy, precision, recall, and F1-scores were chosen to assess the capability of the MLLM to identify the correct disease category. The dataset images were classified into one of two types that included either acquired diseases or inherited diseases. The primary motivation for generalizing disease classes considers the practical scenario of using such a model in a clinical setting. In the current study, only the inherited disease types have additional patient information that could be provided as input to the model, making it difficult to learn enough information to make these types of suggestions.

B. Implementation and Optimization

The proposed MDF-MLLM architecture was developed and trained on a system of 3 nodes containing a total of 12 GPUs. Nodes 1 and 2 each included 4 NVIDIA L40S GPUs and a 64-core Intel Xeon Platinum 8562Y CPU. Node 3 included 4 NVIDIA A40 GPUs and a 32-core Intel Xeon Gold 6326 CPU. The overall model was developed in PyTorch, utilizing an AdamW optimizer and StepLR scheduler (step size = 1, gamma = 0.85) with initial learning rate of 1E-6. Each experimental setup was trained over 60 epochs with a batch size of 2.

For model optimization, binary cross-entropy was utilized for U-Net segmentation loss ($Loss_{Seg}$) and cross-entropy for MLLM token generation loss ($Loss_{MLLM}$). To compensate for any disparity in scale between the two loss functions, Eqn. 7 was utilized to include tunable parameters (λ) that could be manipulated for further optimization, if necessary. For the purposes of this study, λ_{Seg} and λ_{MLLM} were each set to 0.5.

$$Loss_{total} = \lambda_{Seg}Loss_{Seg} + \lambda_{MLLM}Loss_{MLLM} \quad (7)$$

C. Ablation Study and Benchmark Comparison

Due to significant differences in training times between the MLLM and U-Net, each model was trained independently and the pre-trained weights were used to initialize each component of the MDF-MLLM architecture. With the U-Net and MLLM architectures independently trained prior to fusion, it was considered that only the fusion block weights required further tuning to ensure cohesive functionality of the combined model. To properly assess the need for full or partial training, an ablation study was conducted to train MDF-MLLM in various frozen or tunable parameter configurations. Table I demonstrates the performance of MDF-MLLM for each training scenario, with comparison to the respective baseline MLLM.

The MLLM classification accuracy, recall, and F1-score improved considerably as a result of the U-Net fusion, which occurred with both U-Net and MLLM pretrained weights allowed to be further fine-tuned alongside fusion block parameters. This highlights the necessity for all parameters be fine-tuned in a cohesive manner to promote optimal model understanding. When classifying retinal fundus images into disease-related categories, MDF-MLLM improved its classification accuracy

TABLE I: MLLM and fusion model performance by parameter initialization and disease category.

Model	Initialization	Acc.	Disease	Prec.	Rec.	F1
Baseline	MLLM unfrozen	0.60	Acquired	0.95	0.61	0.74
			Inherited	1.00	0.58	0.73
Fusion	U-Net frozen MLLM frozen	0.37	Acquired	1.00	0.47	0.64
			Inherited	1.00	0.12	0.22
Fusion	U-Net frozen MLLM unfrozen	0.78	Acquired	1.00	0.74	0.85
			Inherited	1.00	0.88	0.93
Fusion	U-Net unfrozen MLLM unfrozen	0.94	Acquired	1.00	0.92	0.96
			Inherited	1.00	0.97	0.99

by 56% over baseline. Recall and F1-scores also improved by as much as 67% and 35%, respectively. All training scenarios had samples that were neither classified as acquired or inherited, but the high precision metrics indicate that the model performed well among those that were detected. Improvements in recall and F1-scores demonstrate how well the fusion model was able to reduce the number of non-disease classifications.

A further consideration surrounds the fact that acquired and inherited categories were sourced from different repositories which had no overlapping classes. This poses a question regarding whether the model actually learned to identify relevant disease biomarkers for classification or other nuances (e.g., field of view, lighting, image quality, text sequence length, etc.) that simply helped it to distinguish between data sources. To verify this, disease categories were expanded into specific conditions and detection rates were determined to assess the models capability for within-category classification. Table II displays results for specific disease conditions and Fig. 4 shows the corresponding confusion matrix.

As with the disease categories, the fusion model demonstrated superior performance over baseline by achieving a 119% improvement in overall accuracy, particularly among samples where no diagnosis was provided (None condition). Furthermore, the fusion model demonstrated significant improvement

TABLE II: MLLM and Fusion model performance by disease condition based on the fully trained architecture.

Disease Condition	Baseline MLLM			Fusion MDF-MLLM		
	Prec.	Rec.	F1	Prec.	Rec.	F1
ARMD	1.00	0.02	0.04	0.60	0.06	0.11
Glaucoma	0.44	0.56	0.49	0.50	0.77	0.61
DR	0.35	0.23	0.28	0.49	0.54	0.51
AD RP	0.50	0.07	0.12	1.00	0.64	0.78
AR Stargardt Dis.	0.33	0.25	0.29	1.00	0.75	0.86
Best Disease	0.40	0.31	0.35	0.92	0.85	0.88
AR RP	0.18	0.12	0.14	0.71	0.88	0.79
XL RP	0.78	0.41	0.54	0.84	0.94	0.89
Overall Accuracy	0.26			0.57		

Disease Condition Classification (Baseline model)											
True Labels	AMD	1	16	9	0	0	0	0	0	24	
	GLC	0	29	11	0	0	0	0	0	12	
	DR	0	16	12	0	0	0	0	0	24	
	AD RP	0	2	1	1	2	1	2	1	4	
	AA-STGD	0	0	0	0	2	3	1	0	2	
	BD	0	0	0	0	1	4	1	0	7	
	AR RP	0	3	1	0	1	1	2	1	8	
	XL RP	0	0	0	1	0	1	5	7	3	
	None	0	0	0	0	0	0	0	0	0	
	Predicted labels										
AMD	GLC	DR	AD RP	BD	AR RP	XL RP	None				
Disease Condition Classification (Fusion model)											
True Labels	AMD	3	23	21	0	0	0	0	0	0	3
	GLC	0	40	8	0	0	0	0	0	0	0
	DR	2	17	28	0	0	0	0	0	0	5
	AD RP	0	0	0	9	0	0	3	1	1	
	AA-STGD	0	0	0	0	6	1	1	0	0	
	BD	0	0	0	0	0	11	2	0	0	
	AR RP	0	0	0	0	0	0	15	2	0	
	XL RP	0	0	0	0	0	0	0	16	1	
	None	0	0	0	0	0	0	0	0	0	
	Predicted labels										
AMD	GLC	DR	AD RP	BD	AR RP	XL RP	None				

		Disease Condition Classification (Fusion model)											True Labels
		AMD	GLC	DR	AD RP	AA-STGD	BD	AR RP	XL RP	None			
True Labels	AMD	3	23	21	0	0	0	0	0	0	3		
	GLC	0	40	8	0	0	0	0	0	0	4		
	DR	2	17	26	0	0	0	0	0	0	5		
	AD RP	0	0	0	9	0	0	3	1	1			
	AA-STGD	0	0	0	0	6	1	1	0	0			
	BD	0	0	0	0	0	11	2	0	0			
	AR RP	0	0	0	0	0	0	15	2	0			
	XL RP	0	0	0	0	0	0	0	16	1			
	None	0	0	0	0	0	0	0	0	0	0		
			AMD	GLC	DR	AD RP	AA-STGD	BD	AR RP	XL RP	None		
Predicted labels													

Fig. 4: Confusion matrices for specific disease prediction capabilities on baseline MLLM (left) and U-Net Fused MLLM (right). None classifications represent scenarios in which the MLLM did not mention any disease type in the generated text.

in classification of inherited disease conditions. The acquired conditions also improved, but significant difficulty remained in distinguishing specific types within this category. It is believed that this is likely due to the lack of associated text available for these images. This highlights the importance of having quality text information to provide adequate context to image scenes.

Overall, both within and between disease categories demonstrated improved detection capabilities as a result of injecting the fine-grained pixel-wise U-Net information. The low overall classification accuracy among acquired diseases, however, suggests that the model in its present form would best be used as a screening tool to identify patients in need of a more detailed evaluation. It should be noted that despite the use of pretrained baseline weights during initialization, differences in training speeds were still observed between each sub-architecture of MDF-MLLM, which highlights an important performance trade-off when optimizing such a model for a specific application.

V. CONCLUSION

Retinal fundus photography is a critical diagnostic tool used in ophthalmic health care, and the industry is rapidly adopting technological advances that seek to improve the speed and quality of clinical diagnoses. This research established a novel model architecture (MDF-MLLM) that enhances the classification capability for retinal fundus images into disease categories. MDF-MLLM incorporates global contextual learning collected from image and text information using the LLaMA 3.2 MLLM and fuse it with low-level semantic features extracted from a U-Net architecture, effectively creating a cross-modal learning environment. Notably, MDF-MLLM saw significant increases of 56% and 119% improvement in disease category and disease condition classification accuracies, respectively, compared to the base MLLM on benchmark datasets.

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